MATH 2241 INTRODUCTION TO MATHEMATICAL ANALYSIS, FALL SEMESTER 2025
CHAPTER 3: LIMITS AND CONTINUITY (PART 1)

K.Y. CHAN

Goal of this section:

(1) Cluster points
(2) Limits of functions
(3) Sequential criteria
(4) Limit at infinity
(5) Left and right hand limit

1. Cluster points and Limits of functions

We first discuss the idea of cluster points.

1.1. Cluster of points.

Definition 1.1. Let $A \subset \mathbb{R}$. A point $c \in \mathbb{R}$ is said to be a cluster point of A if for each $\delta > 0$, there exists at least a point $x \in A$ with $x \neq c$ such that $|x - c| < \delta$.

Example 1.2. (1) Let $A = [1, 3]$. Then any point in A is a cluster point of A.
(2) Let $A = \{3, 3.1, 6, 7\}$. Then no point in $\mathbb{R}$ is a cluster point of A.
(3) Let $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$. Then 0 is a cluster point of A, and indeed there is no other cluster points (why?).

Exercise 1.3. Let A be a finite subset of $\mathbb{R}$. How many cluster points in A?

Theorem 1.4. *Let $A \subset \mathbb{R}$. A point c in $\mathbb{R}$ is a cluster point of A if and only if there exists a sequence (a_n) in A such that $\lim_{n\to\infty} a_n = c$ and $a_n \neq c$ for all $n \in \mathbb{N}$.*

Proof. $\Rightarrow$). Let c be a cluster point of A. Then for each $n \in \mathbb{N}$, there exists $a_n \in A$ such that $0 < |a_n - c| < \frac{1}{n}$. Hence, $\lim |a_n - c| = 0$ and so $\lim\ a_n = c$.

$\Leftarrow$). Let (a_n) be a sequence in A such that $\lim_{n\to\infty} a_n = c$ and $a_n \neq c$ for all $n \in \mathbb{N}$. Then, for any $\delta > 0$, there exists $N \in \mathbb{N}$ such that for $n > N$, $|a_n - c| < \delta$ and so in particular, $|a_{N+1} - c| < \delta$. Hence, c is a cluster point of A. □

1.2. **Limit of functions.**

Definition 1.5. Let $A \subset \mathbb{R}$, and let c be a cluster point of A. Let $f : A \to \mathbb{R}$ be a function. We say that L is the limit of f at c if for any given $\epsilon > 0$, there exists a $\delta > 0$ such that for any $x \in A$ satisfying $0 < |x - c| < \delta$, $|f(x) - L| < \epsilon$.

We remark and emphasis that a cluster point c is not necessarily in A.

Exercise 1.6. Draw a curve $y = x^2$. Take $c = 1$. Indicate the region $(c - \delta, c + \delta)$ in x-axis and $(L - \epsilon, L + \epsilon)$ in y-axis to understand the meaning of Definition **??**.

Example 1.7. Let $b \in \mathbb{R}$. Use $\epsilon - \delta$ definition to find the limit:

$$\lim_{x \to c} b = b.$$

(Here we regard b as a constant function $f(x) = b$.)
Let $\epsilon > 0$. We pick δ to be any positive number, say $\delta = 1$. Then for any $x \in \mathbb{R}$ satisfying $0 < |x - c| < \delta = 1$, $|f(x) - b| = |b - b| = 0 < \epsilon$. Then $\lim_{x \to c} b = b$.

Example 1.8. Use the $\epsilon - \delta$ definition to find the limit of

$$f(x) = \begin{cases} 1 & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Then $\lim_{x \to 0} f = 1$.

Exercise 1.9. Use $\epsilon - \delta$ definition to find

$$\lim_{x \to 1} \sqrt[3]{x}$$

Exercise 1.10. Let $f : \mathbb{R} \to \mathbb{R}$ as follows: $f(x) = x$ if x is rational and $f(x) = 0$ if x is irrational. Find the limit of f at $x = 0$ if it exists.

2. The uniqueness and boundedness of a limit

2.1. **Uniqueness of a limit.**

Theorem 2.1. *(Uniqueness of limit) Let $f : A \to \mathbb{R}$. Let c be a cluster point of A. Then there exists at most one limit of f at c.*

Proof. Let L and L' be limits of f at c. Suppose $L \neq L'$ to derive a contradiction. Thus, pick $\epsilon = |L - L'| > 0$. For any $\epsilon > 0$, there exists $\delta > 0$ such that for any $x \in A$ with $0 < |x - c| < \delta$,

$$|f(x) - L| < \frac{\epsilon}{2}.$$

Similarly, there exists $\delta' > 0$ such that for any $x \in A$ with $0 < |x - c| < \delta'$,

$$|f(x) - L'| < \frac{\epsilon}{2}.$$

Take some x^* such that $0 < |x - c| < \min\{\delta, \delta'\}$. But

$$|L - L'| = |L - f(x) + f(x) - L'| \leq |L - f(x)| + |f(x) - L'| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

This gives a contradiction. □

2.2. **Boundedness of limits.**

Theorem 2.2. *(Boundedness of limits) Let $A \subset \mathbb{R}$. Let $f : A \to \mathbb{R}$. If f has a limit at c, then f is bounded on some set*

$$\{x \in A : |x - c| < \delta\}.$$

Proof. Let $L = \lim_{x \to c} f$. Then, choose $\epsilon = 1$. There exists $\delta > 0$ such that, for any $0 < |x - c| < \epsilon$ (with $x \neq c$), $|f(x) - L| < 1$ and so

$$L - 1 < f(x) < L + 1.$$

Then, for any $0 < |x - c| < \epsilon$,

$$|f(x)| < |L| + 1.$$

Then, $|f(x)| < \sup\{|L| + 1, |f(c)|\}$. □

Example 2.3.
(1) Let $f(x) = x^2$. Then f is bounded on $(-\delta + c, \delta + c)$ by $(\delta + |c|)^2$.
(2) Let $f(x) = \frac{1}{x}$ defined for $x \neq 0$. Note that f does not have a limit at 0, and f is not bounded for any $0 < |x| < \delta$.

Exercise 2.4. Is the converse of Theorem **??** true?

3. Sequential criterion for limits

3.1. **Sequential criterion.**

Theorem 3.1. *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then the following statements are equivalent:*

(i) $\lim_{x \to c} f(x) = L$;
(ii) *For any sequence (x_n) in $A - \{c\}$ converging to c, the sequence $(f(x_n))$ converges to L.*

Proof. (i)⇒(ii):
Step 1: Obtain the δ
Suppose $\lim_{x \to c} f(x) = L$. Let (x_n) be a sequence in $A - \{c\}$ which converges to c. Let $\epsilon > 0$. Then there exists $\delta > 0$ such that if $0 < |x - c| < \delta$,

$$|f(x) - L| < \epsilon \tag{3.1}$$

Step 2: Find sufficiently large N so that x_n $(n > N)$ lies in δ-neighborhood
On the other hand, since (x_n) is convergent to c and $x_n \neq c$ for all $n \in \mathbb{N}$, there exists $N \in \mathbb{N}$ such that for any $n > N$,

$$0 < |x_n - c| < \delta$$

Step 3: Combine two conditions
Now, combining the above two assertions, for any $n > N$,

$$|f(x_n) - L| < \epsilon.$$

By definition, $(f(x_n))$ forms a sequence converging to L.

(ii)⇒(i): Suppose (i) does not hold and we shall derive a contradiction.
Step 1: Divergent condition
There exists $\epsilon > 0$ such that for any $\delta > 0$, there exists $x(\delta) \in A$ satisfying $0 < |x(\delta) - c| < \delta$ such that $|f(x(\delta)) - L| \geq \epsilon$.

Step 2: Find a sequence (x_n):
Since c is a cluster point, for each $n \in \mathbb{N}$, we choose a sequence $x_n = x(\frac{1}{n})$. We now show that (x_n) is a sequence converging to c, but $\{f(x_n)\}$ does not converge to $f(c)$.

Step 3: Show $\lim x_n = c$
We first show that (x_n) is a sequence converging to c. Since

$$|x_n - c| < \frac{1}{n},$$

we have $\lim_{n\to\infty} |x_n - c| = 0$ and so $\lim_{n\to\infty} x_n = c$.

Step 4: Show $(f(x_n))$ does not converge to L
Since $|f(x_n) - L| \geq \epsilon$ for all n, $(f(x_n))$ does not converge to L. □

3.2. **Reformulate to a divergence criteria.**

Exercise 3.2. Let $A \subset \mathbb{R}$. Let $f : A \to \mathbb{R}$. Let c be a cluster point of A, and let $L \in \mathbb{R}$. Then f does not have a limit L at c if and only if there exists a sequence (x_n) in $A - \{c\}$ such that $\lim_{n\to\infty} x_n = c$ but $(f(x_n))$ does not converge to L.

Example 3.3. Show that $\sin\left(\frac{1}{x}\right)$ (defined for $x \neq 0$) does not have the limit at 0.
Proof:

(1) Let $x_n = \frac{1}{n\pi}$. Then $(x_n) \to 0$ and $\sin(\frac{1}{x_n}) = \sin(n\pi) = 0$.

(2) Let $y_n = \frac{1}{2n\pi+\frac{1}{2}\pi}$. Then $(y_n) \to 0$ and $\sin(\frac{1}{y_n}) = \sin(2n\pi + \frac{1}{2}\pi) = 1$.

By the sequential criteria (Exercise **??**), f does not have a limit.

Exercise 3.4. Let

$$f(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Show that f does not have a limit at 0.

4. Basic methods on finding limits

4.1. **Basic operations.**

Definition 4.1. Let $A \subset \mathbb{R}$. Let $f, g : A \to \mathbb{R}$. Define

(1) (Sum) $(f+g)(x) = f(x) + g(x)$ for all $x \in A$;
(2) (Difference) $(f-g)(x) = f(x) - g(x)$ for all $x \in A$;
(3) (Product) $(fg)(x) = f(x)g(x)$ for all $x \in A$;
(4) (Multiple) Let $b \in \mathbb{R}$. Then $(bf)(x) = b(f(x))$ for all $x \in A$;
(5) (Quotient) If $g(x) \neq 0$ for all $x \in A$, then define $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$.

Theorem 4.2. *Let c be a cluster point of A. Suppose $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$. Then*

(1) *(Sum)* $\lim_{x\to c}(f+g) = L + M$
(2) *(Difference)* $\lim_{x\to c}(f-g) = L - M$
(3) *(Product)* $\lim_{x\to c}(fg) = LM$
(4) *(Multiple)* $\lim_{x\to c}(bf) = bL$
(5) *(Quotient) If $g(x) \neq 0$ for all $x \in A$ and $L' \neq 0$, then* $\lim_{x\to c} \frac{f}{g} = \frac{L}{L'}$.

Proof. Let (x_n) be a sequence in $A - \{c\}$ converging to c. By disucssions of sequences in previous chapter, we have:

$$\lim_{n\to\infty}(f(x_n) + g(x_n)) = \lim_{n\to\infty} f(x_n) + \lim_{n\to\infty} f(y_n) = L + M.$$

By Theorem **??**, $\lim_{x\to c}(f+g) = f(c) + g(c) = (f+g)(c)$.

Other parts are similar by using the sequential criterion. □

Exercise 4.3. Let $f_1, f_2, \ldots, f_r : A \to \mathbb{R}$. Let $L_1 = \lim_{x\to c} f_1, \ldots, L_r = \lim_{x\to c} f_r$. Then,

(1) for $a_1, \ldots, a_r \in \mathbb{R}$, $\lim_{x\to c}(a_1 f_1 + \ldots + a_r f_r) = a_1 L_1 + \ldots + a_r L_r$.
(2) $\lim_{x\to c}(f_1 \ldots f_r) = L_1 \ldots L_r$.
(3) Use the above to show that if $p(x)$ is a polynomial, $\lim_{x\to c} p(x) = p(c)$.

Example 4.4. Find $\lim_{x\to c} \frac{4x^2-1}{2x-1}$ for all $c \in \mathbb{R}$.

Example 4.5. See Examples 4.2.5 in Bartle-Sherbert book.

4.2. **Squeeze Theorem.** Another useful theorem is the squeeze theorem.

Theorem 4.6. *(Squeeze theorem) Let $A \subset \mathbb{R}$. Let $f, g, h : A \to \mathbb{R}$, and let c be a cluster point of A. If*

$$f(x) \leq g(x) \leq h(x)$$

for all $x \in A$ with $x \neq c$, and $\lim_{x\to c} f(x) = \lim_{x\to c} h(x) = L$ for some $L \in \mathbb{R}$, then $\lim_{x\to c} g(x) = L$.

Proof. Exercise: Use the sequential criteria to prove that. □

Example 4.7. Use the squeeze theorem and $\lim_{x\to 0} x = 0$ to show that: Define $f : [0, \infty) \to \mathbb{R}$ by $f(x) = x^{\frac{3}{2}}$. Then $\lim_{x\to 0} f = 0$.

Exercise 4.8. (1) $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$

(2) By using the inequalities: for $x > 0$,

$$x - \frac{1}{6}x^3 \leq \sin\ x \leq x$$

and for $x < 0$,

$$x - \frac{1}{6}x^3 \geq \sin\ x \geq x,$$

find $\lim_{x\to 0} \frac{\sin\ x}{x}$.

5. One-sided limits

Definition 5.1. Let $A \subset \mathbb{R}$ and let $f : A \to \mathbb{R}$.

(1) Let c be a cluster point of $A \cap (c, \infty)$. We say that L is right hand limit of f at c, written as

$$\lim_{x\to c+} f = L$$

if for any given $\epsilon > 0$, there exists $\delta > 0$ such that for all $x \in A$ with $0 < x - c < \delta$, then

$$0 < |f(x) - L| < \epsilon.$$

(2) Let c be a cluster point of $A \cap (-\infty, c)$. We say that L is left hand limit of f at c, written as

$$\lim_{x\to c-} f = L$$

if for any given $\epsilon > 0$, there exists $\delta > 0$ such that for all $x \in A$ with $0 < c - x < \delta$, then

$$0 < |f(x) - L| < \epsilon.$$

Theorem 5.2. *Let $f : A \to \mathbb{R}$. Let c be a cluster point in both the sets $A \cap (c, \infty)$ and $A \cap (-\infty, c)$. Then $\lim_{x\to c} f = L$ if and only if $\lim_{x\to c+} f = \lim_{x\to c-} f = L$.*

Example 5.3. Let

$$f(x) = \operatorname{sgn}(x) = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$$

Then, $\lim_{x\to 0^+} f = 1$ and $\lim_{x\to 0^-} f = -1$. Hence, $\lim_{x\to 0} f$ does not exist.

Exercise 5.4. Let $f(x) = e^{1/x}$ defined for $x \neq 0$. Use the inequality, for $x > 0$,

$$0 < x < e^x,$$

find $\lim_{x\to 0^+} f$ and $\lim_{x\to 0^-} f$ (if exists). Determine $\lim_{x\to 0} f$.

6. Limit at infinity

Definition 6.1. Let $A \subset \mathbb{R}$ and let $f : A \to \mathbb{R}$. Suppose $(a, \infty) \subset \mathbb{R}$ for some $a \in \mathbb{R}$. We write $\lim_{x\to\infty} f = L$ for some $L \in \mathbb{R}$ if for any $\epsilon > 0$, there exists $K > 0$ such that for any $x > K$, $|f(x) - L| < \epsilon$.

Exercise 6.2. Formulate a definition of $\lim_{x\to-\infty} f(x)$.

Example 6.3. Let $f(x) = \frac{1}{x}$ defined for $x \neq 0$. Then $\lim_{x\to\infty} f = 0$. (*Exercise:* Write a detailed argument for a proof.)

Exercise 6.4. (Sequential Criterion) Show that: $\lim_{x\to\infty} f(x) = L$ if and only if for any sequence (x_n) in A such that $\lim\ x_n = \infty$, $\lim_{n\to\infty} f(x_n) = L$.